

BQ-CA-IMDA: BIGQUERY CONVERSATIONAL ANALYTICS GUIDE

Step-by-Step Instructions for Connecting Gemini in BigQuery (Data Agents, Data Canvas & Conversational Analytics API) to the Extracted SAP Document AI Dataset

DATASET: `sap_docai_staging.extracted_sap_documents` | **SEMANTIC VIEW + DATA AGENT** | **NATURAL LANGUAGE TO SQL**

Target Audience: Google Cloud Partners, Data Engineers, BI Architects, and SAP Business Analysts

1. Executive Summary & Architecture Flow

Once the `docai_sap_pipeline` finishes extracting your 5,000 SAP Invoices (**INVOICE**) and Customer Purchase Orders (**PURCHASE_ORDER**) (25,000 pages/week), all structured header fields, financial totals, and raw JSON entities land in **Google Cloud BigQuery**:

- **Dataset:** `${GCP_PROJECT_ID}.sap_docai_staging`
- **Base Table:** `${GCP_PROJECT_ID}.sap_docai_staging.extracted_sap_documents`

This **BQ-CA-IMDA** guide provides the step-by-step instructions to tie that extracted BigQuery dataset into **BigQuery Conversational Analytics** (powered by **Gemini in BigQuery** and the **Conversational Analytics API**), enabling finance, procurement, and operations users to ask natural-language questions over live SAP documents and automatically generate verified SQL queries, KPI summaries, and visual charts.

End-to-End Data & Conversational Analytics Flow

Stage	Google Cloud Component	Target Resource / Artifact
Stage 1: Extraction	Document AI Batch Pipeline (<code>docai_sap_pipeline</code>)	Streams 5,000 extracted SAP PDFs into <code>\${GCP_PROJECT_ID}.sap_docai_staging.extracted_sap_documents</code> .
Stage 2: Semantic Enrichment	BigQuery DDL Metadata & Flattened View	Attaches SAP DDIC descriptions to columns and creates <code>sap_docai_staging.vw_sap_conversational_analytics</code> unpacking <code>raw_entities_json</code> .
Stage 3: Conversational Agent	BigQuery Conversational Analytics (Data Agent)	Configures the <code>sap-docai-conversational-agent</code> with SAP system instructions, business rules, and Golden SQL examples.
Stage 4: Business Consumption	BigQuery Data Canvas & Conversational Analytics API	Business users chat in natural language via BigQuery Studio UI or REST API (<code>geminidataanalytics.googleapis.com</code>).

2. Environment Setup & Enabling Conversational Analytics APIs

Step 2.1: Export Your Environment Variables

Run this block in your terminal (using the same `GCP_PROJECT_ID` and `GCP_ACCOUNT_EMAIL` from your Document AI deployment):

```
export GCP_PROJECT_ID="YOUR_GCP_PROJECT_ID"
export GCP_ACCOUNT_EMAIL="YOUR_GCP_ACCOUNT_EMAIL"
export GCP_BQ_LOCATION="US"
export GCP_BQ_DATASET="sap_docai_staging"
export GCP_BQ_TABLE="extracted_sap_documents"
```

Step 2.2: Enable BigQuery Conversational Analytics & Gemini APIs

Run the following **gcloud** command to enable the Conversational Analytics (**geminidataanalytics**), Gemini for Google Cloud (**cloudaicompanion**), and Dataplex metadata APIs:

```
gcloud services enable \
geminidataanalytics.googleapis.com \
cloudaicompanion.googleapis.com \
bigquery.googleapis.com \
dataplex.googleapis.com \
--project="${GCP_PROJECT_ID}"
```

Step 2.3: Grant Required Conversational Analytics IAM Roles

Grant your user account (or partner service account) the IAM roles required to create and query Conversational Analytics Data Agents and BigQuery Data Canvas:

```
for ROLE in \
roles/geminidataanalytics.dataAgentCreator \
roles/geminidataanalytics.dataAgentUser \
roles/cloudaicompanion.user \
roles/bigquery.admin; do
gcloud projects add-iam-policy-binding "${GCP_PROJECT_ID}" \
--member="user:${GCP_ACCOUNT_EMAIL}" \
--role="${ROLE}" \
--quiet
done
```

3. Enriching the BigQuery Dataset for High-Accuracy Text-to-SQL

BigQuery Conversational Analytics achieves its highest SQL accuracy when:

1. **Table and column descriptions** clearly explain SAP technical codes (**LIFNR**, **KUNNR**, **NETWR**, **MWSKZ**, **BUKRS**).
2. **Nested JSON strings** (**raw_entities_json**) are exposed as typed SQL columns in a dedicated **Semantic View**.

Step 3.1: Verify Extracted Rows Exist in BigQuery

Before configuring Conversational Analytics, confirm your **extracted_sap_documents** table has rows populated:

```
bq query --use_legacy_sql=false \
"SELECT document_type, COUNT(*) AS total_docs, ROUND(SUM(total_amount), 2) AS
total_value_sgd
FROM \`${GCP_PROJECT_ID}.${GCP_BQ_DATASET}.${GCP_BQ_TABLE}`\`
GROUP BY document_type"
```

Step 3.2: Add SAP Business Descriptions to the Base Table Columns

Run this **bq query** command to attach rich SAP metadata descriptions directly onto **\${GCP_PROJECT_ID}.sap_docai_staging.extracted_sap_documents**:

```
bq query --use_legacy_sql=false "
ALTER TABLE \`${GCP_PROJECT_ID}.${GCP_BQ_DATASET}.${GCP_BQ_TABLE}\`
SET OPTIONS (
description = 'SAP Document AI extracted Vendor Invoices (MIRO/VBRK) and Customer Purchase
Orders (VA01/VBAK). Contains 1 row per multi-page PDF document.'
);

ALTER TABLE \`${GCP_PROJECT_ID}.${GCP_BQ_DATASET}.${GCP_BQ_TABLE}\`
ALTER COLUMN document_id SET OPTIONS (description = 'Unique PDF filename identifier (e.g.
SAP_INV_20260001_5p or SAP_PO_20260002_5p)'),
ALTER COLUMN document_type SET OPTIONS (description = 'SAP Document Category: INVOICE
(Vendor Invoice) or PURCHASE_ORDER (Customer PO)'),
ALTER COLUMN sap_document_number SET OPTIONS (description = 'Official SAP Billing / PO
Document Number (BELNR / VBELN / EBELN, e.g. INV-2026-900001 or PO-4500900002)'),
ALTER COLUMN sap_vendor_lifnr SET OPTIONS (description = 'SAP 10-digit Vendor Account Number
(LFA1-LIFNR, e.g. 0000100881)'),
ALTER COLUMN sap_customer_kunnr SET OPTIONS (description = 'SAP 10-digit Customer Sold-To
Party Number (KNA1-KUNNR, e.g. 0000201102)'),
ALTER COLUMN document_date SET OPTIONS (description = 'Document Issue Date (BLDAT / AUDAT)
in YYYY-MM-DD format'),
ALTER COLUMN due_date SET OPTIONS (description = 'Payment Due Date / Net Maturity Date
(ZFBDT) in YYYY-MM-DD format'),
ALTER COLUMN currency SET OPTIONS (description = 'SAP Transaction Currency Key (WAERK,
default SGD)'),
ALTER COLUMN net_amount SET OPTIONS (description = 'SAP Net Document Value excluding tax
(NETWR) in transaction currency'),
ALTER COLUMN tax_amount SET OPTIONS (description = 'SAP GST / Tax Amount (WMWST, 9%
Singapore GST code G9)'),
ALTER COLUMN total_amount SET OPTIONS (description = 'Gross Total Payable Amount including
tax (WRBTR /Gross Value = net_amount + tax_amount)'),
ALTER COLUMN confidence_score SET OPTIONS (description = 'Google Cloud Document AI
extraction confidence score between 0.0 and 1.0 (e.g. 0.985 = 98.5%));"
```

Step 3.3: Create the Enriched Conversational Analytics View (vw_sap_conversational_analytics)

Run the following command to create a flattened, business-ready view that parses **raw_entities_json** (**sap_mandt**, **sap_bukrs**, **MWSKZ** tax code, **processor_type**), casts dates to native **DATE** types, and maps SAP Vendor/Customer IDs to human-readable enterprise names:

```

bq query --use_legacy_sql=false "
CREATE OR REPLACE VIEW
  \`${GCP_PROJECT_ID}.${GCP_BQ_DATASET}.vw_sap_conversational_analytics\`
OPTIONS(
  description = 'Curated SAP Document AI Semantic View for BigQuery Conversational Analytics.
    Unpacks JSON fields, casts dates, and maps SAP LIFNR/KUNNR master data names.'
) AS
SELECT
  document_id,
  document_type,
  sap_document_number,
  sap_vendor_lifnr,
  CASE sap_vendor_lifnr
  WHEN '0000100881' THEN 'ASEAN Precision Manufacturing Pte Ltd'
  WHEN '0000103910' THEN 'Global Cloud Logistics (SG) Ltd'
  WHEN '0000108421' THEN 'Pacific Semiconductor Supply Co'
  WHEN '0000109200' THEN 'Merlion Industrial Chemicals Pte Ltd'
  ELSE 'Other SAP Vendor (' || sap_vendor_lifnr || ')'
  END AS vendor_name,
  sap_customer_kunnr,
  CASE sap_customer_kunnr
  WHEN '0000201102' THEN 'Suntec Enterprise Holdings'
  WHEN '0000200891' THEN 'Jurong Port Operations Corp'
  WHEN '0000204519' THEN 'Changi Aviation Services Ltd'
  WHEN '0000209940' THEN 'Marina Bay Retail Group'
  ELSE 'Other SAP Customer (' || sap_customer_kunnr || ')'
  END AS customer_name,
  SAFE.PARSE_DATE('%Y-%m-%d', document_date) AS document_date,
  SAFE.PARSE_DATE('%Y-%m-%d', due_date) AS due_date,
  DATE_DIFF(SAFE.PARSE_DATE('%Y-%m-%d', due_date), SAFE.PARSE_DATE('%Y-%m-%d', document_date),
    DAY) AS payment_terms_days,
  currency,
  ROUND(net_amount, 2) AS net_amount,
  ROUND(tax_amount, 2) AS tax_amount,
  ROUND(total_amount, 2) AS total_amount,
  ROUND(confidence_score * 100, 2) AS confidence_pct,
  JSON_VALUE(raw_entities_json, '$.sap_mandt') AS sap_client_mandt,
  JSON_VALUE(raw_entities_json, '$.sap_bukrs') AS sap_company_code_bukrs,
  JSON_VALUE(raw_entities_json, '$.extracted_fields.MWSKZ') AS sap_tax_code_mwskz,
  JSON_VALUE(raw_entities_json, '$.processor_type') AS docai_processor_type,
  source_gcs_uri,
  processed_timestamp
FROM \`${GCP_PROJECT_ID}.${GCP_BQ_DATASET}.${GCP_BQ_TABLE}\`;

```

4. Configuring BigQuery Conversational Analytics in Google Cloud Console

Option A: Create a Dedicated Conversational Analytics Data Agent (Recommended)

A **Data Agent** allows you to pre-configure the SAP tables, system instructions, and Golden SQL rules once so any authorized business user can chat with the SAP dataset.

1. Open BigQuery Studio:

Navigate to <https://console.cloud.google.com/bigquery> and select project `${GCP_PROJECT_ID}`.

2. Open Conversational Analytics (Agents):

In the left navigation bar (or top BigQuery Studio toolbar), click **Agents** (or **Conversational Analytics** -> **Create Data Agent**).

3. Configure Agent Identity:

- **Agent Name:** SAP Document AI Financial & Procurement Assistant
- **Agent ID:** sap-docai-conversational-agent
- **Description:** Conversational AI assistant for querying extracted SAP Vendor Invoices (MIRO/VBRK) and Customer Purchase Orders (VA01/VBAK) in sap_docai_staging.

4. Attach BigQuery Data Sources:

Click **Add Data Source** and select both:

- `${GCP_PROJECT_ID}.sap_docai_staging.vw_sap_conversational_analytics` (Primary semantic view)
- `${GCP_PROJECT_ID}.sap_docai_staging.extracted_sap_documents` (Base extraction table)

5. Paste System Instructions & SAP Semantic Rules:

In the **System Instructions / Context** field of the Data Agent, copy and paste the following grounding block:

You are an expert SAP Financial & Procurement Data Analyst querying Google Cloud BigQuery dataset `sap\_docai\_staging`.

Always prefer querying the view `sap\_docai\_staging.vw\_sap\_conversational\_analytics` because it includes mapped `vendor\_name`, `customer\_name`, parsed `DATE` columns (`document\_date`, `due\_date`), and unpacked SAP JSON fields (`sap\_company\_code\_bukrs`, `sap\_tax\_code\_mwskz`).

Mandatory SAP Business & SQL Rules:

1. Document Types (`document\_type`):
 - Use `document\_type = 'INVOICE'` for SAP Accounts Payable / Vendor Billing Invoices (SAP T-Code MIRO / Table VBRK).
 - Use `document\_type = 'PURCHASE\_ORDER'` for Customer / Procurement Purchase Orders (SAP T-Code VA01 / Table VBAK).
2. Financial Amounts:
 - `net\_amount` is SAP `NETWR` (amount before GST tax).
 - `tax\_amount` is 9% Singapore GST (`MWSKZ = 'G9'`).
 - `total\_amount` is Gross Payable Amount (`net\_amount + tax\_amount`). Always specify `SGD` when reporting currency totals.
3. SAP Master Data Codes:
 - Always display both the 10-digit SAP code (`sap\_vendor\_lifnr` or `sap\_customer\_kunnr`) AND the human-readable name (`vendor\_name` or `customer\_name`).
4. Quality & Audit Filtering:
 - `confidence\_pct` represents the Document AI extraction confidence percentage (e.g., 98.5%). Flag documents with `confidence\_pct < 98.0` if asked about manual review or audit exceptions.

6. Save & Publish:

Click **Publish** (or **Save**). You can now open the chat pane directly inside BigQuery Studio and ask questions in plain English!

Option B: Interactive Exploration in BigQuery Data Canvas

1. In the BigQuery Explorer pane, expand `${GCP_PROJECT_ID}` -> `sap_docai_staging`.
2. Right-click `vw_sap_conversational_analytics` and select **Query in Data Canvas**.
3. In the **Natural Language Prompt** bar below the table node, type your question (for example: "Compare total net amount and tax amount by vendor_name for INVOICE documents and plot a bar chart").
4. Click **Generate** -> Click **Run** -> Click **Visualize** to auto-generate interactive bar, line, or pie charts alongside the SQL query.

5. Programmatic Setup & Chat via Conversational Analytics REST API

You can also create the Data Agent and send conversational queries programmatically via the **Google Cloud Conversational Analytics API** (geminidataanalytics.googleapis.com), which powers custom web portals and drives the BigQuery Conversation Analytics token SKU.

Step 5.1: Create the Data Agent via REST API (curl)

```
ACCESS_TOKEN="$(gcloud auth print-access-token)"

curl -s -X POST \
  "https://geminidataanalytics.googleapis.com/v1alpha/projects/${GCP_PROJECT_ID}/locations/global/dataAgents?data_agent_id=sap-docai-agent" \
  -H "Authorization: Bearer ${ACCESS_TOKEN}" \
  -H "Content-Type: application/json" \
  -d "{
    \"description\": \"SAP Document AI Invoice and PO Conversational Agent\",
    \"data_analytics_agent\": {
      \"published_context\": {
        \"system_instruction\": \"Query sap_docai_staging.vw_sap_conversational_analytics. Filter document_type='INVOICE' for invoices and 'PURCHASE_ORDER' for POs. Always report currency in SGD.\",
        \"datasource_references\": {
          \"bq\": {
            \"table_references\": [
              {
                \"project_id\": \"${GCP_PROJECT_ID}\",
                \"dataset_id\": \"sap_docai_staging\",
                \"table_id\": \"vw_sap_conversational_analytics\"
              }
            ]
          }
        }
      }
    }
  }"
```

Step 5.2: Send a Natural Language Question to the Conversational Analytics :chat Endpoint

```
curl -s -X POST \
  "https://geminidataanalytics.googleapis.com/v1alpha/projects/${GCP_PROJECT_ID}/locations/global:chat" \
  -H "Authorization: Bearer ${ACCESS_TOKEN}" \
  -H "Content-Type: application/json" \
  -d "{
    \"parent\": \"projects/${GCP_PROJECT_ID}/locations/global\",
    \"messages\": [
      {
        \"user_message\": {
          \"text\": \"What is the total gross invoice amount, net amount, and GST tax amount grouped by vendor_name?\"
        }
      }
    ],
    \"data_agent_context\": {
      \"data_agent\": \"projects/${GCP_PROJECT_ID}/locations/global/dataAgents/sap-docai-agent\"
    }
  }"
```

6. Curated Partner Demo Prompts & Verified Golden SQL Queries

Use these 6 high-impact natural-language prompts during customer/partner workshops in **BigQuery Conversational Analytics** or **Data Canvas**:

#	Natural Language Prompt to Ask Gemini	What Conversational Analytics Generates & Returns
1	"Show total Invoice spend (net, 9% GST tax, and gross total in SGD) broken down by SAP Vendor LIFNR and Vendor Name."	Filters document_type = 'INVOICE' , groups by sap_vendor_lifnr , vendor_name , and sums net_amount , tax_amount , and total_amount .
2	"Compare total Purchase Order value versus total Invoice value for each SAP Customer KUNNR and Customer Name."	Pivots SUM(CASE WHEN document_type='PURCHASE_ORDER'...) vs SUM(CASE WHEN document_type='INVOICE'...) by customer to highlight PO-to-Invoice billing gaps.
3	"What is our daily document ingestion volume and cumulative gross spend trend over time by document type?"	Groups by document_date , document_type , computes daily COUNT(*) and window function SUM(total_amount) OVER (ORDER BY document_date) , and plots a time-series line chart.
4	"Which top 10 highest-value invoices are due within the next 30 days, and who is the vendor?"	Filters document_type = 'INVOICE' , orders by total_amount DESC LIMIT 10 , showing sap_document_number , vendor_name , due_date , and total_amount .
5	"Give me an audit summary of Document AI extraction confidence scores and list any documents below 98.5% confidence."	Computes AVG(confidence_pct) , MIN(confidence_pct) , and lists documents where confidence_pct < 98.5 along with their source_gcs_uri link in Cloud Storage.
6	"Break down total extracted document value by SAP Company Code (BUKRS), Tax Code (MWSKZ), and Document AI Processor Type."	Unpacks JSON attributes sap_company_code_bukrs , sap_tax_code_mwskz , and docai_processor_type to verify SAP posting compliance (BUKRS = 1000 , MWSKZ = G9).